

Integrating climate risk into the 2025 EU-wide stress test: the effects of climate risks for firms

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As authorities across the euro area work towards including climate risks into regular stress-testing frameworks, this article offers a starting point for assessing bank resilience to climate risks that materialise under a short-term horizon. This is relevant since acute physical risks and abrupt policy change can also materialise at short notice and affect the balance sheet of financial institutions. The analysis uses an adverse macroeconomic backdrop that combines the EBA's adverse scenario with the Network for Greening the Financial System's Nationally Determined Contributions (NGFS NDCs) scenarios. It extends the EU-wide 2025 stress test results by incorporating both transition and acute physical climate risks into the credit risk assessment for non-financial corporations by means of top-down models. Transition risks driven by green investments to reduce emissions amplify credit losses and reduce banks' Common Equity Tier 1 (CET1) capital, particularly in high energy-intensive sectors. Similarly, acute physical risks, such as extreme flood events, reduce CET1 capital through direct damage, local disruptions, and macroeconomic spillovers. While the magnitude of impacts varies across banks, the analysis shows that both types of climate risk can have a moderate but consequential effect on capital ratios. Notably, the banks most exposed to climate-related losses may differ from those identified as the most vulnerable in the broader EU-wide assessment. These findings underscore the importance of incorporating both types of climate risk into regular financial stability assessments.

1 Introduction

In recent years integrating short and long-term climate risks into stress testing has emerged as a key priority for financial regulators globally. Forward-looking assessment methods have become a crucial instrument for quantifying and assessing the potential impacts of climate change on economies and financial systems.^[1] While much attention has been placed on the long-term nature of these risks, their short-term implications are equally important. Acute weather events, abrupt policy shifts and rapid market repricing driven by climate developments can occur at short notice, causing immediate and significant impacts on financial institutions' balance sheets and the broader economy.

Financial regulators in Europe will be adding climate risk monitoring to their regular stress testing of the financial sector. The European Supervisory Authorities (the European Banking

Authority (EBA), the European Insurance and Occupational Pensions Authority (EIOPA) and the European Securities and Markets Authority (ESMA) – collectively referred to as the ESAs) recently issued a [Joint Consultation Paper](#) on draft guidelines on the stress testing of environmental, social and governance (ESG) risks. This started with the environmental component, focusing on climate and other nature-related risks such as biodiversity, deforestation, etc. Financial institutions face challenges in modelling future climate pathways, including fragmented tools and a lack of modelling consensus; these draft guidelines aim to harmonise methodologies and practices among banking, insurance and securities supervisors to ensure proportionality and enhance the effectiveness and efficiency of stress tests. The consultation process concluded on 19 September 2025, and the final guidance is expected to be published by the ESAs in the first few months of 2026.

With regards to the banking system, the EBA is working to integrate climate risks into its EU-wide stress-testing framework. Based on the strategy outlined in the EBA's [Annual Report 2024](#) and in line with its mandate,^[2] the incorporation of climate risks into the EU-wide stress-testing framework will be gradual. Partial integration of climate risks, referred to as a combined approach, will start in 2027, with additional related elements introduced in subsequent stress tests. Particular emphasis will be placed on ensuring comprehensive coverage and assessment of both physical risks (including their acute dimensions) and transition risks, supported by the development of tailored scenarios.

The EBA's proposed framework for climate stress testing will ensure that the principles of proportionality and simplification are observed while leveraging the existing EU-wide stress test infrastructure. The proportionality principle means the framework will be tailored to the size, risk profile and climate risk exposure of individual institutions. Moreover, the climate stress-testing module will be aligned with the EU-wide stress test in terms of data definitions, reporting processes, scenarios and methodological design, significantly reducing complexity and easing the implementation burden for institutions. Finally, utilising the existing EU-wide stress test infrastructure will enhance consistency and efficiency, while also paving the way for the gradual and comprehensive integration of climate risks into the EU-wide framework over the longer term.

This article provides additional insights into the EU-wide stress test by incorporating climate risk into banks' credit risk projections via a top-down approach. The analysis extends the 2025 EU-wide stress test results by incorporating both transition and acute physical climate risks into the credit risk assessment for non-financial corporations (NFCs) by means of top-down models. The focus on credit risk is justified as (1) it is a significant risk driver in supervisory stress test exercises; and (2) the transmission channels from climate shock to credit risk are better understood and, to varying degrees, better reflected in banks' credit risk and stress test models compared with other risk drivers such as market or profitability risks. Transition risks stemming from green investments to reduce emissions increase default probabilities, particularly in high energy-intensive sectors. This leads to amplified credit losses, lowering banks' CET1 capital by a moderate 74 basis points. Extreme flood events further exacerbate credit risks through direct, local and macroeconomic transmission channels, resulting in an additional 77 basis point decrease in the CET1 ratio (see Box 1). Interestingly, the exercise identifies undetected pockets of risk, as the banks most exposed to climate-related losses may differ from those identified as the most vulnerable in the broader EU-wide assessment (see [Rodríguez d'Acuña and Shaw, 2025](#)).

An integrated approach to climate stress tests, when executed well, can help banks perform better, not just meet regulatory expectations. While climate stress-testing exercises have emerged as a key tool for supervisors to assess the impact of climate risks on the banking system, euro area banks themselves are making more and more use of them to inform required disclosures and strategic choices (see [ECB report on good practices for climate stress testing](#)). In view of the evolving nature of this topic, banks will have to adapt their practices on an ongoing basis.

2 Extending the 2025 EU-wide stress test by adding a climate risk analysis for NFCs

Banks' sectoral credit risk exposures to transition risks can be translated into changes in default probabilities and credit losses. The analysis leverages the projections for energy prices, emissions and energy consumption, as well as the trajectories of relevant macroeconomic variables provided by the Network of Central Banks and Supervisors for Greening the Financial System (NGFS) in its Nationally Determined Contributions (NDCs) scenario. By integrating the NGFS NDCs scenario into the EBA's 2025 adverse scenario, this approach provides insights into how transition risks can amplify the outcomes of the official EU-wide stress test, highlighting their importance for financial stability assessments. The impact of transition risk must be assessed in conjunction with physical risk (see **Box 1**), as delayed and fragmented transition policies are associated with higher physical risk over the medium term.

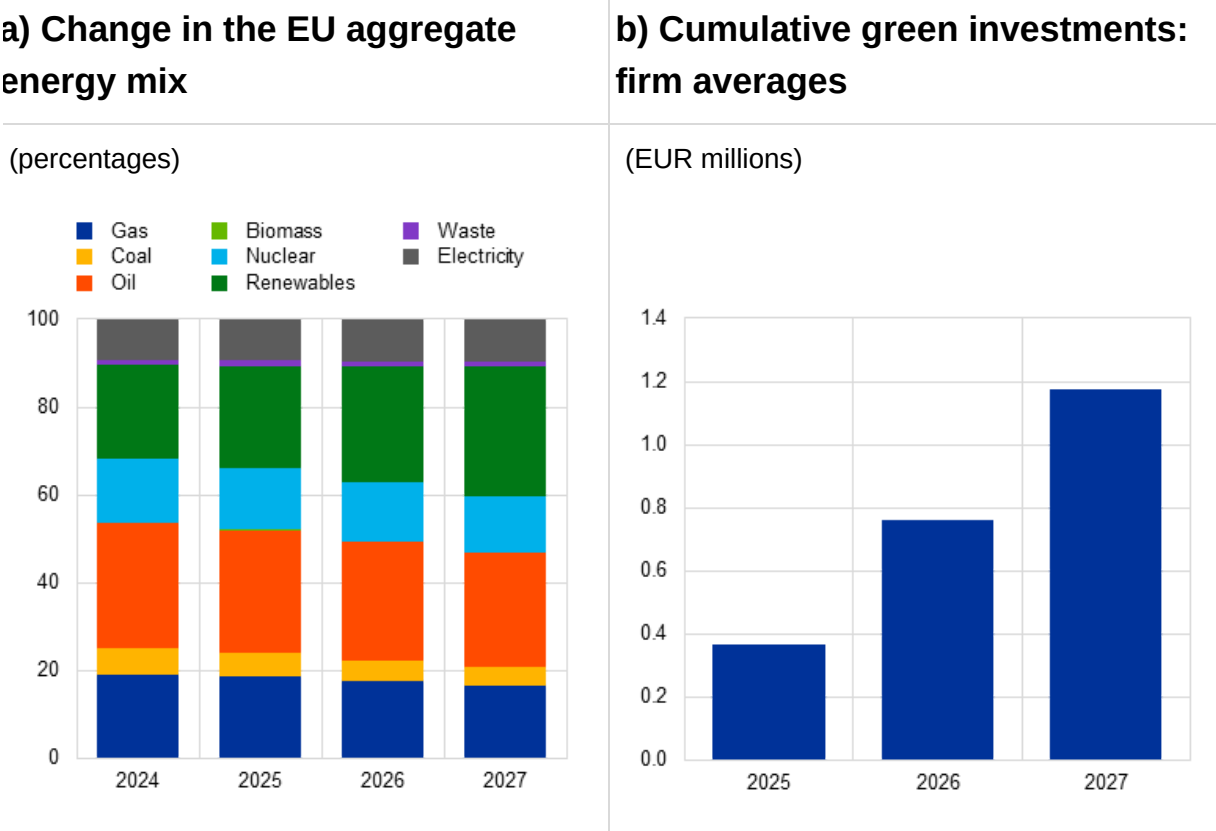
2.1 Transition risk scenario

Building on the NGFS NDCs scenario, a short-term transition risk scenario is constructed, reflecting pledged emissions targets and delayed policy action. While climate risk scenarios are typically designed with a long-term horizon, transition risks can also be highly relevant in the short term. This article adopts a short-term perspective on climate transition risk using the 2025-27 path of the NGFS NDCs scenario. This incorporates all pledged emission reduction targets announced by individual countries as of March 2024, even where they have not yet been accompanied by effective policies.^[3] Specifically, the NGFS NDCs scenario foresees a shift in the EU energy mix, marked by a reduction in the consumption of fossil fuels such as gas and coal and growing reliance on renewables and electricity, driven by firm-level green investments. Out to 2027, the share of gas, coal and oil in the energy mix is projected to decrease, while renewables are expected to experience a substantial increase, reflecting the transition towards cleaner energy sources (**Chart 1**, panel a).

To understand how these changes affect individual firms, the energy mix foreseen by the scenario is downscaled at firm level. This is done by combining data on sectoral energy consumption from Eurostat with the revenue share of each firm within its sector. Furthermore, it is assumed that the technological change needed to support the implied shift in firms' energy mix is achieved through green investments (**Chart 1**, panel b), with the magnitude of these directly proportional to the emission intensity of each sector. Green investments affect firms' leverage and profitability, and hence their probability of default.

Chart 1

The energy mix and green investments implied by the NGFS NDCs scenario



Sources: ECB and ECB calculations.
Notes: EU aggregate figures. Panel a) shows the share of energy type by year. Panel b) shows average cumulative green investments for EU firms over the stress test horizon 2025-27.

2.2 Methodological approach to projecting default probabilities and loan losses under the transition risk scenario

To project corporate default probabilities over the stress test horizon, a fixed effects sector-level regression is employed linking corporate failure rates to leverage and profitability. NFCs' probabilities of default (PDs) are estimated by leveraging the approach employed by the ECB in the Fit-for-55 scenario analysis, with some adjustments (see **Appendix I** in ESAs and ECB, 2024). Annual sector-specific PDs are projected by linking sectoral failure rates to projected profitability and leverage, both of which incorporate exogenous climate-related shocks as defined by the NGFS NDCs scenario. The failure rate at sectoral level is calculated as the percentage of failed firms relative to all firms in the sector. Corporate failure is measured by an indicator variable adapted from Gourinchas et al. (2024). This indicator takes a value of 1 if the firm's interest expenses exceed its cash holdings and its leverage is larger than 1 over two consecutive years, and 0 otherwise. Leverage is defined as the ratio of total liabilities to total assets, and profitability as the ratio of revenue net of operating expenses to total assets.

Profitability is influenced by several interrelated factors, including interest expenses and the amortisation cost associated with green investments. Investments aimed at reducing CO₂ emissions are assumed to be paid off and amortised over a ten-year period. Firms' profitability is reduced by interest expenses on the outstanding amount and the amortisation costs of green investments. Additionally, as firms' total assets are projected in line with the NGFS scenario path for macroeconomic variables, the transition to a low-carbon economy has an impact on both the numerator and the denominator of leverage, negatively affecting solvency.^[4]

Increased green investments, as outlined in the NGFS scenario, affect firms' balance sheets through higher indebtedness and lower profitability. The required level of green investment is assumed to depend on the energy intensity and emissions associated with the sector in which a firm operates. Mining and quarrying, manufacturing, electricity generation and transmission and water supply are examples of high energy-intensive sectors. Building, transport and agriculture are considered medium energy-intensive sectors. And some wholesale and trade activities, accommodation and food services, ICT and others are classified as low energy-intensive sectors.

To quantify the effect of climate transition risks on credit losses, projected changes in sectoral PDs are mapped to banks' sectoral exposures. AnaCredit data are used to map the projected changes in PD at sectoral level to firms' PD and their creditor bank. Thereafter the change in the PDs of NFCs is aggregated at country level and applied to a bank's initial PD reported in the EU-wide stress test. Combining the resulting credit losses with those reported under the EBA's 2025 adverse scenario makes it possible to assess how transition risks can amplify the credit losses in an adverse macroeconomic scenario.

2.3 Transition risk results

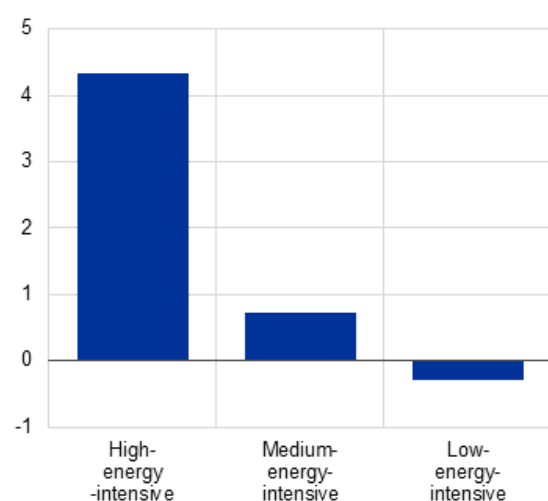
The leverage and profitability of high energy-intensive firms react most strongly to the transition risk shocks included in the scenario. The average cumulative increase in leverage relative to the 2024 starting point is most pronounced for high energy-intensive firms. They rise by 4.33 percentage points, compared with an increase of 0.74 percentage points in the medium energy-intensive sectors and a decrease of 0.30 percentage points in the low energy-intensive sectors (**Chart 2**, panel a). Similarly, profitability declines most in the sectors that are heavily reliant on carbon-intensive energy, with an average cumulative reduction of 8.41 percentage points for high energy-intensity sectors and 1.96 percentage points for the others (**Chart 2**, panel b). These dynamics are driven by several factors. The pronounced decline in profitability observed in energy-intensive sectors reflects not only the conservative assumptions used to translate the scenario variables into firm-level balance sheet outcomes, but also the added burden of higher amortisation and interest expenses associated with green investments. Funds raised to finance green investments also contribute to a significant increase in leverage, adding further strain to firms' finances. At the same time, investments allow firms to reduce their consumption of carbon-intensive energy, which lowers their energy costs.

Chart 2

Change in leverage and profitability, by energy intensity

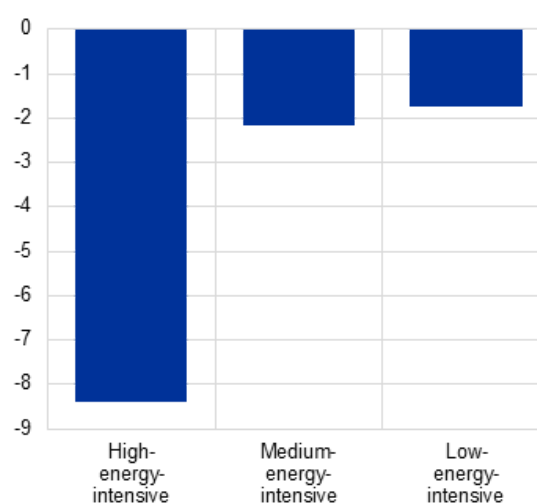
a) Leverage

(percentage point change)



b) Profitability

(percentage point change)

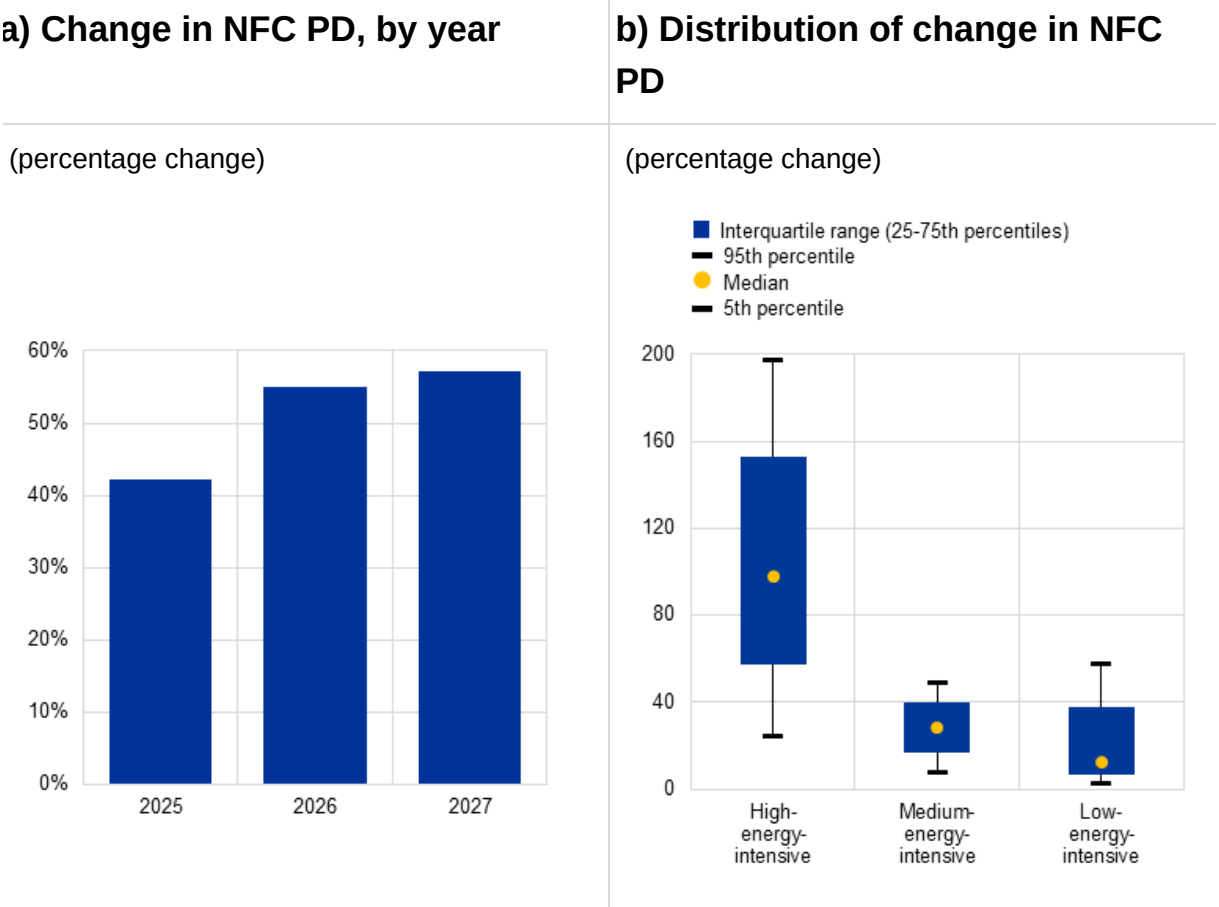


Sources: ECB, BvD Orbis and ECB calculations.

Note: The figures represent the cumulative percentage point change in leverage (panel a) and profitability (panel b) from 2024 to 2027 under the combined EBA and NGFS scenario at EU aggregate level.

The increase in PD over the three-year horizon is more pronounced for firms operating in high energy-intensive sectors. On average, firms' PDs rise by 50%, with the largest share of the increase concentrated in the first year of the stressed period (**Chart 3**, panel a). Sectors with high energy intensity face the most substantial increase, with default probabilities climbing by a median of 91%, while firms in medium energy-intensity sectors show a more moderate median rise of 28% (**Chart 3**, panel b).

Chart 3
Firms' probability of default



Sources: ECB, BvD Orbis and ECB calculations.
Notes: Panel a) shows the cumulative percentage change in NFC PD by year. Panel b) shows the distribution of the cumulative percentage change in NFC PD from 2024 to 2027 across different sectors by energy intensity.

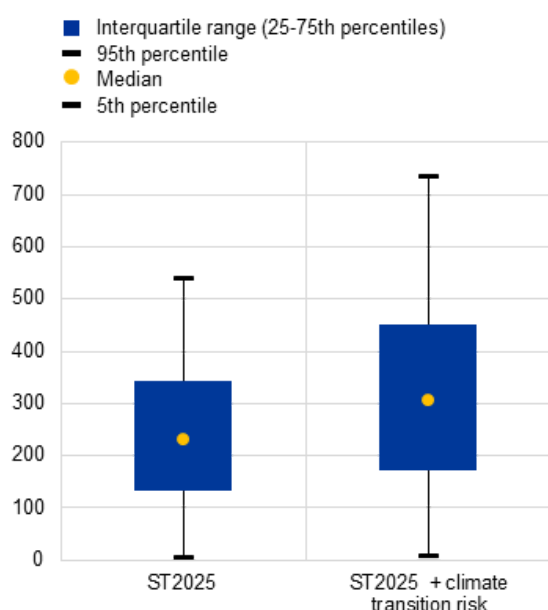
The overall impact of loan losses on banks' CET1 ratios remains contained; however, banks with higher exposures to energy-intensive sectors face greater losses. Compared with the 2025 EU-wide stress test results, the additional system-level impact of credit risk losses from banks' NFC portfolios on their CET1 capital ratios under the adverse scenario remains moderate, accounting for 74 basis points over the 2025-27 horizon (**Chart 4**, panel a). The largest additional credit risk losses arise for banks that are most exposed to high energy-intensive sectors, followed by those with the highest exposures to medium and low energy-intensive sectors (**Chart 4**, panel b).

Chart 4

Distribution of NFC credit risk losses and impact on CET1 capital ratios

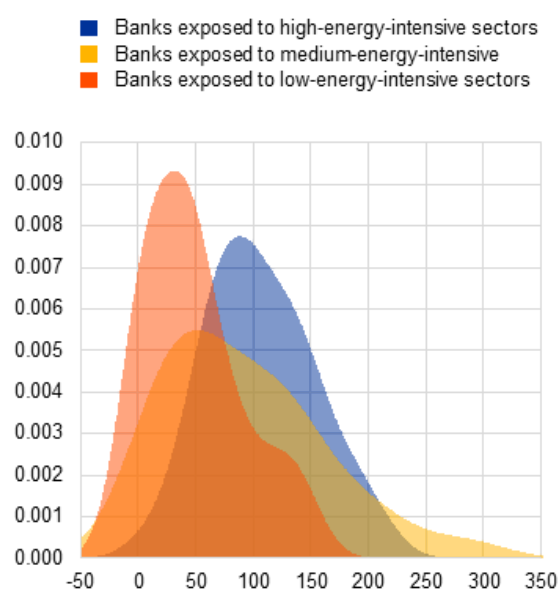
a) NFC loan losses in the EU-wide stress test with additional climate transition risk

(basis points)



b) CET1 impact of additional NFC loan losses due to climate transition risk in the adverse scenario, by bank exposure

(x-axis: basis points, y-axis: density)



Sources: 2025 EU-wide stress test and ECB calculations.

Notes: Panel a) shows the distribution of NFC credit risk losses under the EBA's adverse scenario with additional transition risk shocks. ST2025 stands for the 2025 EU-wide stress test. Panel b) shows a kernel density of the distribution of the incremental loan losses of transition risk on NFCs under the EBA's adverse scenario by exposure to sectors with different energy intensity. Banks are ranked by exposure to high, medium and low energy-intensive sectors. The blue, yellow and orange areas show the distribution of the impact for banks with the highest exposure to each of these in order.

Box 1

Flood events in the EU and impact on corporate loan quality

Prepared by Aurora Abbondanza, Ugo Albertazzi, Davor Djekic and Aurea Ponte Marques

This box presents a sensitivity analysis of the 2025 EU-wide stress test results looking at the effects of acute physical risks on corporate loan quality. The EBA's 2025 adverse scenario captures the main cyclical risks faced by the EU banking sector, but does not

account for climate risk, despite its potential to trigger significant and systemic losses.^[5] In this box, bank resilience is therefore assessed against the stress triggered by the financial and the real-economy shocks underpinning the EBA's adverse scenario in conjunction with the materialisation of acute climate physical risks. The analysis focuses on credit risk, the primary transmission channel through which physical risks affect banks' financial soundness. It concentrates on corporate loans, considering both data availability and the ubiquity of the business practice of ensuring residential mortgage collateral for physical risk. The analysis is limited to flood events, as these are the best documented in terms of their direct impact on firms' activity. However, including other physical climate hazards (e.g. wildfires or droughts) would provide a more comprehensive view of country and sector-specific heterogeneity, as exposure to different types of risks varies widely across countries.

The analysis is based on a scenario combining the EBA adverse and the NGFS NDCs scenarios, making it possible to consider the materialisation of physical risk against a backdrop of significant macroeconomic challenges, including a projected cumulative GDP decline of approximately 6.9 percentage points (see **Section 2** for a detailed description of the NGFS NDCs scenario).

The materialisation of acute physical risk events may lower the credit quality outlook through three distinct transmission channels, each operating at a different level of aggregation. First, floods may have a direct impact on the solvency of the firms directly affected, due to the disruptions and physical asset damage these events typically entail. Second, they may also produce adverse effects on the credit risk of all firms in the area, even those not directly affected, due to disruption to local transportation or service availability, for example. Third, floods may lead to higher credit risk if they result in macroeconomic deterioration, as is the case in the scenario under consideration. This implies that effective stress testing for acute physical risk requires empirical models that capture the credit risk implications of such events at both the firm and local economy levels. It also demands an approach able to construct a meaningful, granular scenario that clearly identifies the specific local areas and firms affected by the acute event assumed to materialise in the scenario.

A credible assessment of the impact of acute physical risk scenarios should be conducted at a granular level, however, to capture their local nature. Physical risk exposure from river flooding is highly concentrated in specific regions and affects a relatively small but significant subset of firms in the euro area. The physical risk score measures each borrower's exposure to river flooding for the period 2021-50 under the RCP 4.5 scenario, ranging from 0 (low risk) to 5 (very high risk). As shown in **Chart A**, panel a), while most firms in the sample exhibit low exposure to physical risk (score 0), a substantial number face moderate to high risk, with more than 22,000 falling into the highest risk category (score 5).

Turning to the empirical evidence, physical climate risks can elevate financial vulnerability across wider regions through macroeconomic channels. Under the EBA's adverse scenario combined with the NGFS NDCs scenario, degradation of the macroeconomic environment leads to a broad-based deterioration in credit quality. The probability of corporate defaults rises significantly, peaking in 2026 and reaching levels

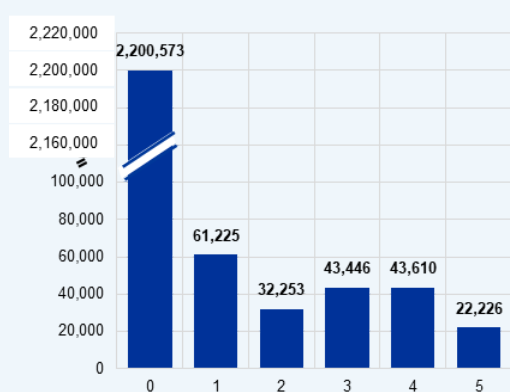
consistently higher than in the EU-wide stress test (**Chart A**, panel b). The implied increase in default frequency is around 2 percentage points, cumulatively at the end of the projection horizon.

Chart A

Probability of default under the EBA's adverse scenario and NGFS NDCs scenario and firm distribution of exposure to physical risk

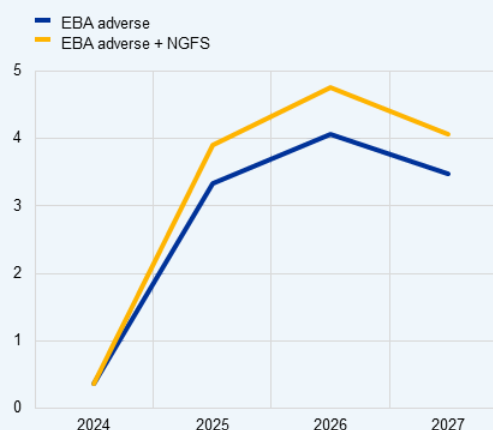
a) Number of firms, by physical risk score

(x-axis: physical risk score; y-axis: number of firms)



b) Probability of default under the EBA adverse and NGFS NDCs scenarios

(percentage, y-axis: probability of default)



Source: ECB calculations.

Notes: The histogram in panel a) refers to firms reported in AnaCredit by at least one bank in the sample of the 2025 EU-wide stress test in December 2024. The physical risk score refers to river flooding and is based on the RCP 4.5 scenario set out by the Intergovernmental Panel on Climate Change for the period 2021-50 and is calculated at the borrower level. This granular indicator was computed as part of the ECB's analytical indicators on physical risk statistics. 0 means low risk, 5 very high risk. Panel b) shows the probability of default from top-down stress test models under the EBA's 2025 adverse scenario and NGFS NDCs scenario.

A granular physical risk impact assessment, consistent with the combined EBA adverse and NGFS NDCs scenario, can be obtained based on the principle that the affected areas and firms are those displaying the highest levels of physical risk score (Chart A, panel a). This impact assessment assumes materialisation of acute physical risk in the form of widespread floods.^[6] The geographical distribution of floods in the scenario is calibrated by assuming that such events concern areas with some exposure to physical risks, that is all municipalities other than those only populated by firms with the lowest physical risk

score.^[7] This approach identifies 2,786 municipalities as affected, representing 36% of loan exposures in the sample (**Chart B**, panel b). Within each affected municipality, the 6.4% of firms with the highest climate risk score are assumed to be affected (**Chart B**, panel a). This fraction corresponds to what has been historically observed in the sample used for the RDD analysis (**Table A**, panel b).

Table A

Effect of being in a flooded area (DiD) and effect of being flooded (RDD) on loan quality

a) Difference-in-difference (DiD) results

(coefficients, standard errors in parenthesis)

	(1)	(2)	(3)
Flooded	-0.002 (0.018)	0.002 (0.023)	0.022 (0.018)
Post	-0.023 (0.023)	-0.011 (0.048)	-0.012 (0.020)
Flooded x Post	0.099 ** (0.042)	0.181 ** (0.074)	0.107 ** (0.053)
Quarters from event	[-2,2]	[-2,2]	[-8,2]
Observations	38,522	33,482	72,900
Number of events	55	25	25

b) Regression discontinuity design (RDD) results

(coefficients, standard errors in parenthesis)

	(1)	(2)	(3)
Flooded	0.720 ** (0.362)	1.967 ** (0.919)	1.885 *** (0.652)
Distance	0.002 *** (0.001)	0.005 *** (0.002)	0.004 *** (0.001)
Flooded x Distance	-0.000 (0.002)	-0.002 (0.004)	-0.003 (0.003)
Quarters from event	(0,1]	(0,1]	(0,2]
Observations	9,705	2,257	4,506
Number of events	6 ^[1]	1 (Valencia)	1 (Valencia)

Source: ECB calculations.

Notes: ***, ** and * denote significance at the 1, 5 and 10 percent levels respectively. Panel a) shows the outcome of a difference-in-difference exercise testing the difference in default probability between firms (regardless of whether or not they are directly affected) in municipalities (NUTS 4) affected by the event and those not affected, before and after the event. The coefficients in bold provide an estimate of the causal effects of being an affected firm on the percentage probability that a performing loan outstanding at the time of the event becomes a deteriorated loan in one of the following quarter(s). Quarterly averages. Panel b) shows the outcome of a regression discontinuity exercise testing the difference in default probability between firms inside and outside the geographical area affected by the event under examination, but arbitrarily close to its border. The coefficients in bold provide an estimate of the causal effects of being an affected (flooded) firm on the percentage probability that a performing loan outstanding at the time of the event becomes a deteriorated loan in one of the following quarter(s). Quarterly averages.

[1] The six events in the RDD sample for which granular geospatial data are available are as follows: flood in southern Ireland (02/2021), flood in Ebro River basin (12/2021), flood in Marche and Umbria regions (09/2022), flood in Emilia-Romagna (05/2023), flood in Tuscany (11/2023), flood in Valencia (10/2024). Some notable events are excluded from the RDD sample due to data limitations but are included in the DiD sample.

The impact of physical risk events on corporate loan quality is significant for firms based in affected municipalities, even if they are not directly affected by the event. A difference-in-difference (DiD) exercise assesses the impact of a wide set of events (up to 55) on the loan quality of firms located in affected areas, regardless of whether or not they were directly flooded. The results show that firms in these municipalities also faced a higher likelihood of loan deterioration after the event, with estimated PD increases ranging from 0.1 to 0.2 percentage points compared with unaffected municipalities over the next two quarters, even after considering overall economic conditions (**Table A**, panel a). These findings are consistent across different timeframes and a wide set of events, confirming the broader regional impact of physical risk, and possibly reflecting disruptions caused to the local economy.

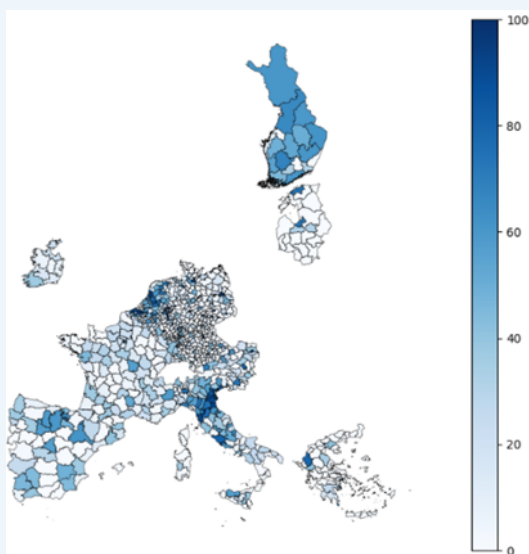
Zooming further in on the differential effect at firm level reveals a clear and significant impact of floods on the corporate loan quality of firms directly affected. Using a regression discontinuity design (RDD) based on detailed geolocation data covering both firm locations and the areas affected by acute events, the analysis compares firms located in affected areas with neighbouring unaffected ones, focusing on six major flood events in the euro area since 2018.^[8] The results indicate that for loans extended to flooded firms, the probability of becoming non-performing increases by 0.7 percentage points in the next quarter compared with unaffected firms (**Table A**, panel b). In the specific case of the Valencia flood in 2024, the effect was even stronger – up to 2.0 percentage points in the next two quarters (**Table A**, panel b, columns 2 and 3). These estimates are statistically significant and suggest a clear causal relationship between direct exposure to physical risk materialisation and worsening loan performance.^[9]

Chart B

Share of affected firms by municipality and shares of firms by type of exposure to physical risk

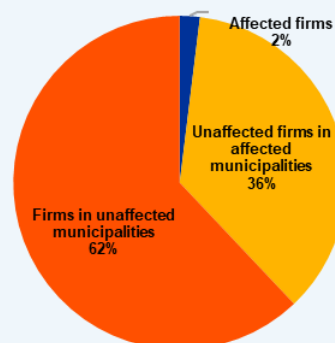
a) Map of municipalities, gradient coloured by share of affected firms in the granular physical risk scenario

(percentage, share of affected firms)



b) Shares of firms by type of exposure to physical risk in the granular physical risk scenario

(percentages)



Source: ECB calculations.

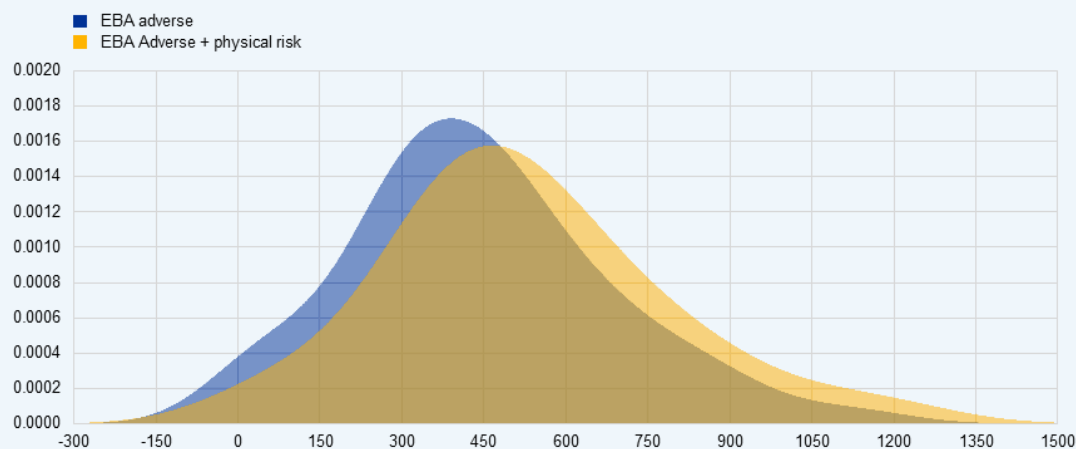
Notes: Panel a) shows the EU map of municipalities, gradient coloured by share of affected firms (white for a low share, dark blue for the highest). Panel b) shows the EU aggregate share of firms by type of exposure to physical risk.

Capital depletion from the granular physical risk scenario is obtained by integrating the macroeconomic effect of the combined EBA and NGFS scenarios and the estimated impact on credit risk for loans to unaffected firms in affected municipalities, directly affected firms and other domestic firms, with bank-level information on the size of such portfolios. Building on AnaCredit information, each bank's portfolio of corporate loans can be divided into the three categories. On aggregate, around 34% of exposures are associated with firms not directly affected by the physical risk scenario but located in affected municipalities. Loans to directly affected firms account for just 2% of banks' total portfolios, with the remaining exposures corresponding to other domestic firms. This breakdown allows us to compute the losses obtained in the physical risk scenario for each bank, based on the quantification provided above.^[10]

Chart C

Distribution of bank credit risk losses under the EBA adverse and physical risk (NGFS NDCs combined with granular) scenarios

(x-axis: basis points, y-axis: density)



Sources: ECB (2025) and ECB calculations.

Note: Kernel density estimate plot visualising the distribution of impairments across banks.

The additional credit risk losses stemming from physical climate risk are relatively contained at the aggregate level, but exhibit significant heterogeneity across banks.

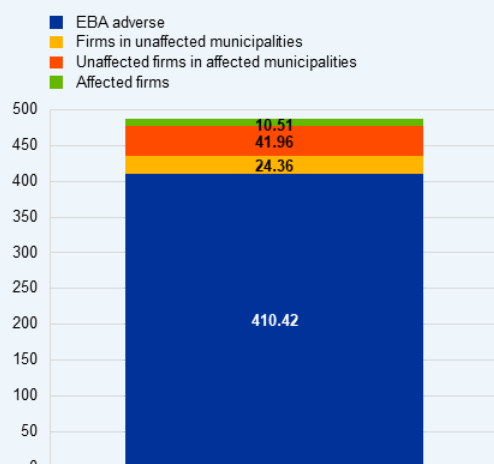
Chart D, panel a) presents the aggregate results, showing that the combined impact of the EBA's adverse scenario and the NGFS NDCs results in an overall capital depletion of 487 basis points – nearly 77 basis points more than the impact of the EBA's adverse scenario alone. This difference reflects the impact of climate risks and is almost entirely driven by local effects. Interestingly, the largest increases in losses are not concentrated among banks already facing substantial capital depletion under the EBA's adverse scenario, indicating a weak relation between the two sources of risk. For the median bank, capital depletion under the combined scenario is 496 basis points, compared with 419 basis points under the EBA's adverse scenario (**Chart C**). Notably, for 7% of banks, losses exceed 200 basis points, indicating a concentration in exposure to physical climate risk. Finally, as shown in **Chart D**, this concentration of losses can arise from any of the three transmission channels through which physical risk materialises: losses from directly affected firms, losses from unaffected firms located in affected areas and losses from broader macroeconomic deterioration.

Chart D

Breakdown of system-level credit risk losses and banks' distribution of climate credit risk losses by type of exposure to physical risk

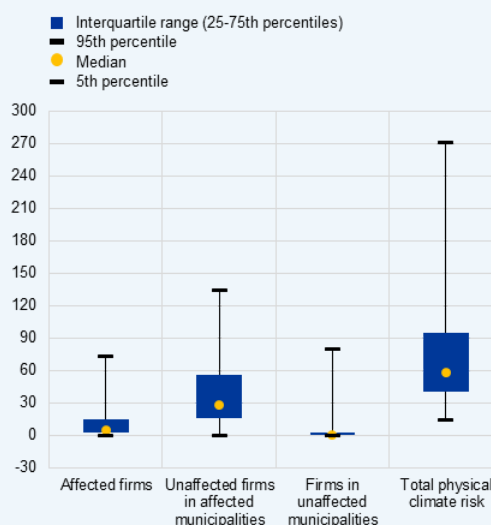
a) Breakdown of system-level credit risk losses, by type of exposure to physical risk

(basis points)



b) Distribution of climate credit risk losses, by type of exposure to physical risk

(basis points)



Sources: EU-wide stress test and ECB calculations.

Notes: Panel a) shows the system-level breakdown of credit risk losses by type of exposure to physical risk, in addition to the 2025 EU-wide stress test credit risk losses. Panel b) shows the distribution of the additional physical climate credit risk losses by type of exposure to physical risk. Note that our sample of banks is slightly smaller than the EBA sample; as a result, the figures on CET1 ratio depletion are slightly different.

3 Conclusions

Analysing transition and physical climate risks reveals a moderate yet consequential impact on bank resilience, highlighting the importance of comprehensive financial risk assessments. As authorities across the euro area work towards including climate risks in regular stress-testing frameworks, this analysis offers a starting point for a sensitivity analysis exploring bank resilience to transition and physical climate risks under an adverse macroeconomic scenario (see EBA, 2024). For transition risk, the analysis shows that the additional impact of transition risk on CET1 capital depletion is moderate on aggregate, at 74 basis points, with the largest part of the impact stemming from exposures to high energy-intensive sectors. The impact of physical risk is assessed in Box 1, looking at

a sensitivity analysis where a lack of, or delayed and fragmented transition, policies is associated with higher physical risk over the medium term. The impact of acute physical risk is estimated to increase CET1 capital depletion by 77 basis points beyond the depletion indicated by the EBA's adverse scenario. Our analysis indicates that transition and physical climate risks can have a moderate but consequential impact on banks' capital ratios. Additionally, the banks most exposed to climate-related losses may differ from those identified as most vulnerable in the broader EU-wide assessment (see [Rodriguez d'Acri and Shaw, 2025](#)). Together, these findings underscore the importance of incorporating both types of climate risk into regular financial stability assessments.

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1.

For an overview of recent developments and practices in climate stress testing, as well as a non-exhaustive list of climate stress tests conducted or underway across financial jurisdictions globally, see [UNEP \(2024\)](#).

2.

Article 23 of the EBA founding regulation requires the EBA to set up an adequate stress testing regime of potential systemic risks “including potential environmental-related systemic risk”.

3.

The NGFS NDCs scenario assumes an increase in global temperature of 2.3°C by the end of the century, a slow pace of technological change and a delayed policy reaction.

4.

While the horizon of short-term NGFS scenarios is more compatible with the exercises presented in this article, the long-term NGFS scenario is preferable overall as it provides a more comprehensive macro-financial characterisation, including crucial energy-related indicators such as energy prices and sectoral energy consumption, which are required to run the ECB's top-down (climate) models. By building on the long-term NGFS scenario, the analysis benefits from a more balanced and consistent set of variables, ensuring greater robustness and comparability in stress-testing applications.

5.

See, for instance, ESRB (2025).

6.

The floods are assumed to occur in the first of year of the projection horizon, in line with the approach followed in the Disasters and Policy Stagnation short-term scenario recently published by the NGFS. The NGFS NDCs scenario does not list explicit assumptions used in acute physical risk modelling.

7.

A materiality threshold is applied to account for potential errors in geospatial data: a municipality is considered affected if it has at least a few affected firms. The results remain robust to variations in this threshold.

8.

Other forms of acute physical risk events that occurred recently in the EU, such as wildfires, turned out not to have affected geographical areas where firms are present. This does not exclude the presence of more indirect and longer-term economic effects on people, ecosystems and assets in vulnerable areas, but assessment of these is outside the scope of this box. See, for example, Oom et al. (2022).

9.

Note that these results implicitly embed the effects of mitigation measures such as moratoria programmes or other fiscal interventions that may have been put in place by local or national governments to support affected firms. This is not an issue provided it can be considered as part of the automatic stabilisers; if so, the estimated impact is a correct representation of what can be expected in a hypothetical acute event, neglecting the adoption of specific policy interventions beyond what is embedded in historical regularities.

10.

Note that both the DiD and RDD estimates represent local average treatment effects for affected municipalities and firms respectively. Consequently, when losses are aggregated at the portfolio level, adjustments are made to account for double counting when needed.

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